# EEZ BB3+ wire harness

This is a wire harness for the EEZ BB3+ with two power modules. For an additional module, it is necessary to include one more *AC power module cable* and one *DC power module cable*.

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## *Cooling fan cable*

| | |
|---|---|
| Total length | 70 (± 5) mm |
| Type | 3 wire, already mounted on the cooling fan |
| Fan: | TME: MF92251V1-G99-A<br>LCSC: C55123210<br>Mouser: 369-MF92251V1100UG99 |



**Connectors**

| **Side A** | X13 | **Side B** | N/A |
|---|---|---|---|
| Housing type | Receptacle, 0.1” | Fan is already equipped with cable. Therefore it only has to be terminated with the connector on side A. | |
| TME | MX-22-01-3037 | | |
| LCSC | C185238 | | |
| Mouser | 538-22-01-3037 | | |



| | |
|---|---|
| Pin | Female |
| Pin quantity | 3 |
| TME | MX-0850-0114 |
| LCSC | C45408619 |
| Mouser | 538-08-50-0114 |



**Wiring**

| Name | Position | Color / description |
|---|---|---|
| GND | 1 | Black |
| +12 Vdc | 2 | Red |
| Tacho | 3 | Yellow |

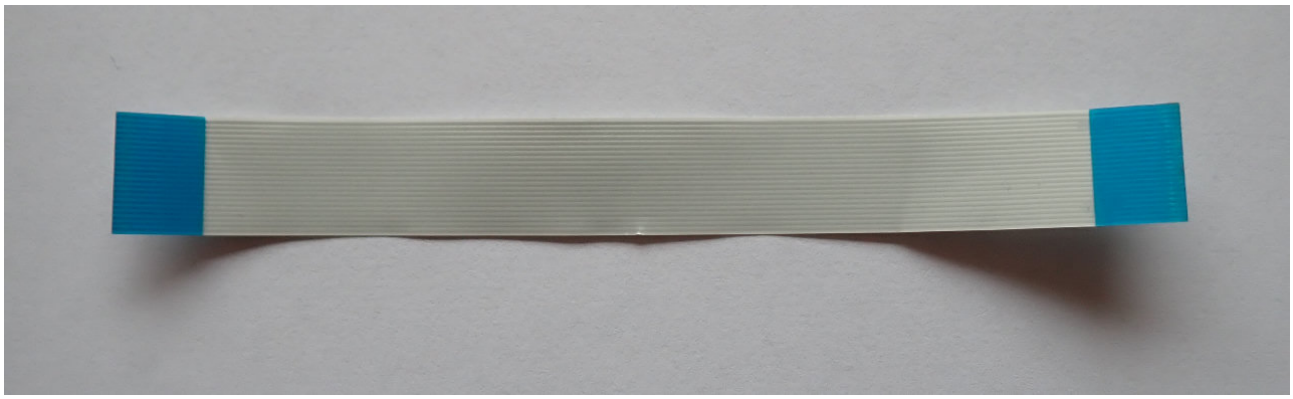
**Picture**



## *TFT module FFC cable*

| | |
|---|---|
| Length | 101 mm, 0.5 mm pitch, 22 pin |
| Type | Assembled ribbon cable with IDC connectors; 16x28AWG |
| TME | 05-22-A-0101-A |
| LCSC | C18024481 |
| Mouser | 640-522A0101A4064T |

**Picture**



## *AC power module cable*

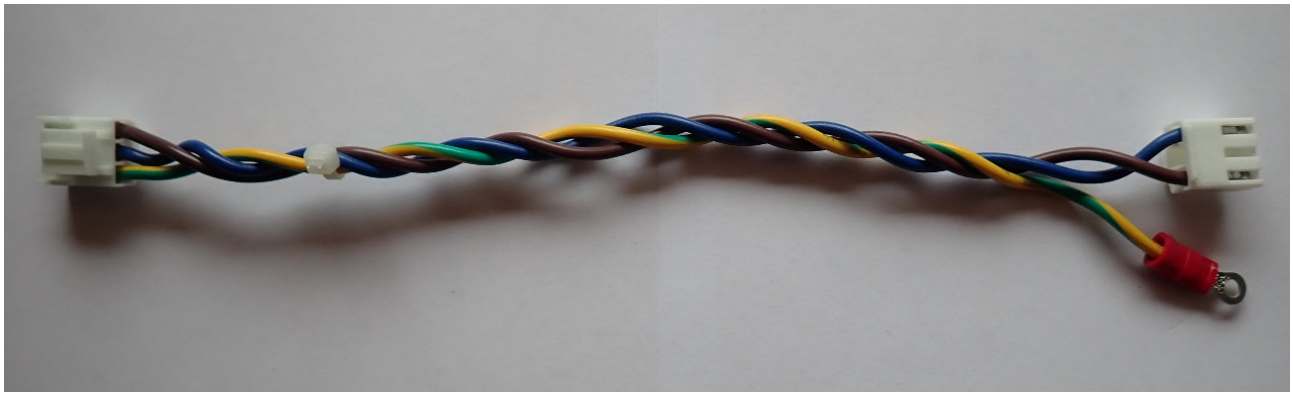
| | |
|---|---|
| Length | 200 (± 5) mm |
| Type | 0.75 $mm^2$ (18 AWG) single wire, stranded, min. 250 VAC |
| TME | Brown: H05VK075BR, Blue: H05VK075BL, Green-Yellow: H05VK075GN/GE |
| LCSC | – |
| Mouser | Brown: 602-3055-100-07, Blue: 602-3055-100-06, Green-yellow: 602-3055-100-28 |

**Connectors**

| **Side A** | Mainboard (X10, X11 or X12) | **Side B** | AC/DC module power output |
|---|---|---|---|
| Housing type | Female, wire-to-board connector, 3.96 mm pitch | Housing type | Female, wire-to-board connector, 3.96 mm pitch |
| TME | VHR-3N | TME | VHR-3N |
| LCSC | C157899 | LCSC | C157899 |
| Mouser | 306-VHR-3N | Mouser | 306-VHR-3N |
| |  | |  |
| Pin | Female | Pin | Female |
| | 3 | Pin quantity | 2 (middle position is not populated!) |
| TME | NS39-T | TME | NS39-T |
| LCSC | C160349 | LCSC | C160349 |

| | | | |
|---|---|---|---|
| Mouser | 306-SVH-21T-P1.1 | Mouser | 306-SVH-21T-P1.1 |
| |  | |  |
| | | Insulated ring crimp terminal | Ring inside diameter 3.2 mm |
| | | TME | – |
| | | LCSC | C20232480 |
| | | Mouser | 548-BU-190540035 |
| | | |  |

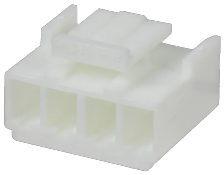
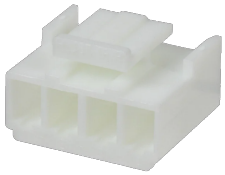
**Picture**



## *DC power module cable*

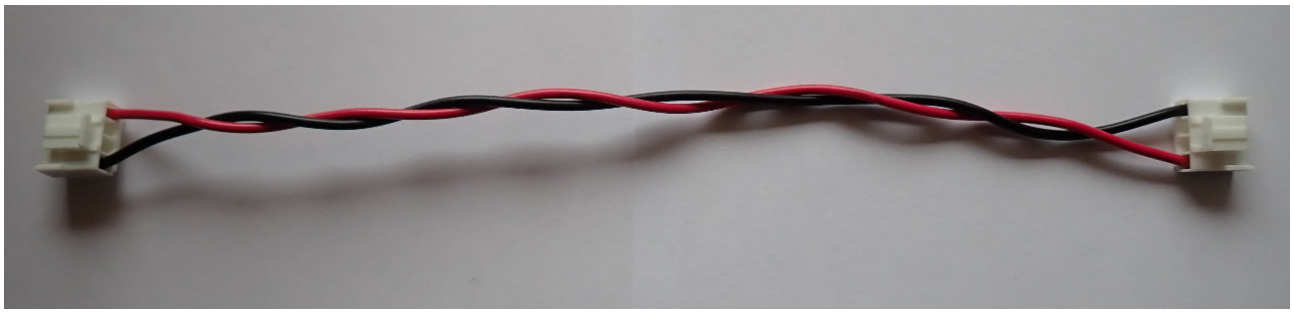
| | |
|---|---|
| Length | 370 (± 5) mm |
| Type | 0.75 $mm^2$ (18 AWG) single wire, stranded, min. 250 VAC |
| TME | Red: H05VK075RT, Black: H05VK075SW |
| LCSC | – |
| Mouser | Red: 602-3055-100-03, Black: 602-3055-100-02 |

**Connectors**

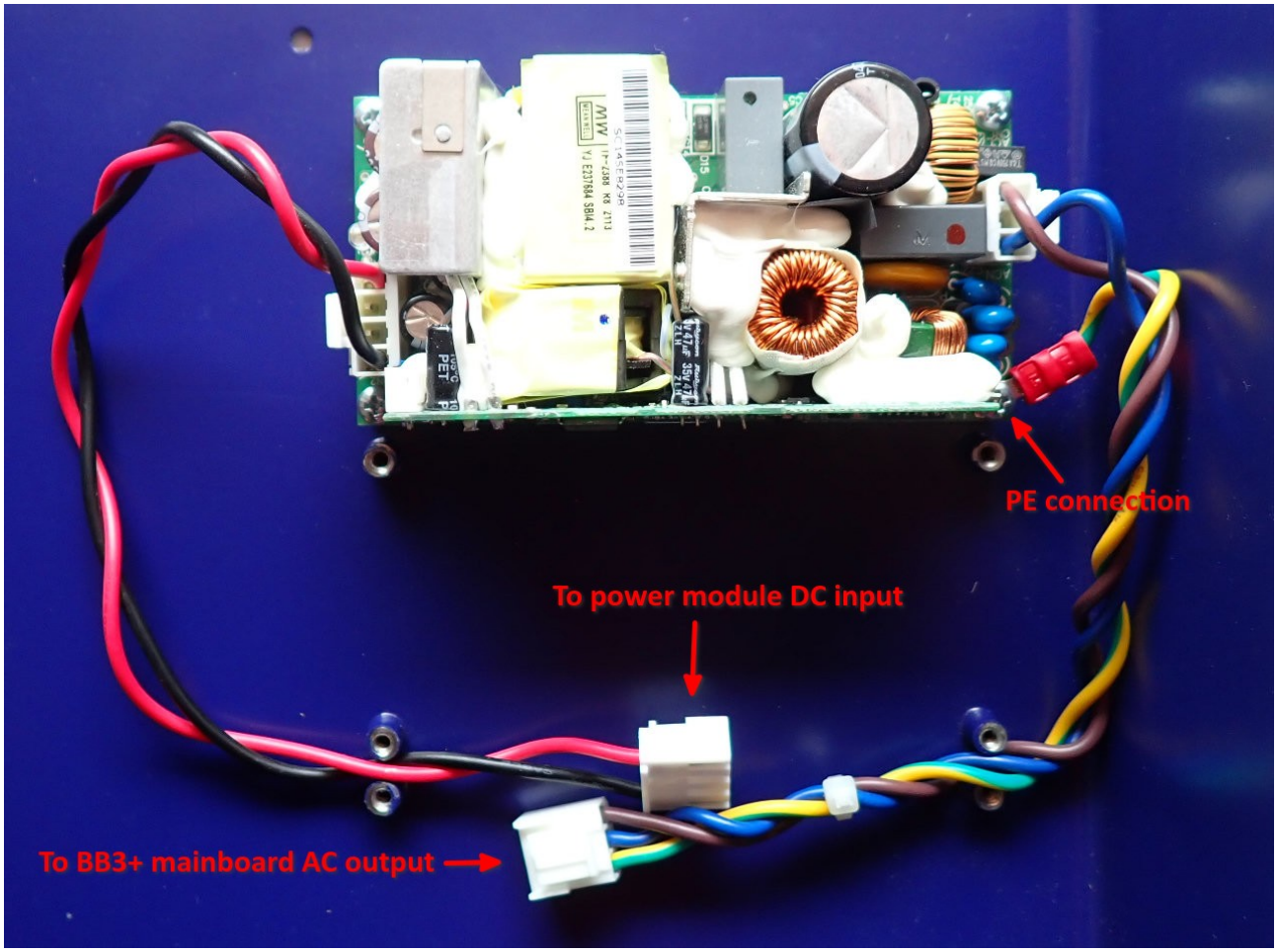
| **Side A** | Power module (X3) | **Side B** | AC/DC module power output |
|---|---|---|---|
| Housing type | Female, wire-to-board connector, 3.96 mm pitch | Housing type | Female, wire-to-board connector, 3.96 mm pitch |
| TME | VHR-4N | TME | VHR-4N |
| LCSC | C157898 | LCSC | C157898 |
| Mouser | 306-VHR-4N | Mouser | 306-VHR-4N |
| |  | |  |

| Pin | Female | Pin | Female |
|---|---|---|---|
| Pin quantity | 2 | Pin quantity | 2 |
| TME | NS39-T | TME | NS39-T |
| LCSC | C160349 | LCSC | C160349 |
| Mouser | 306-SVH-21T-P1.1 | Mouser | 306-SVH-21T-P1.1 |
| |  | |  |

**Picture**



View of the AC/DC module mounted on the BB3+ top plate, with cables for the power module connected:



## BOM

The following BOM includes items for connecting two power modules.

| Type | Part (TME) | Part (LCSC) | Part (Mouser) | Total | |
|---|---|---|---|---|---|
| 3-pin 0.1” receptacle | MX-22-01-3037 | C185238 | 538-22-01-3037 | 1 |  |
| 0.1” pin | MX-0850-0114 | C45408619 | 538-08-50-0114 | 3 |  |
| TFT module FFC cable | 05-22-A-0101-A | C18024481 | 640-522A0101A4064T | 1 | |
| 3-pin header 3.96 mm | VHR-3N | C157899 | 306-VHR-3N | 4 |  |
| 4-pin header 3.96 mm | VHR-4N | C157898 | 306-VHR-4N | 4 |  |
| 3.96 mm pin | NS39-T | C160349 | 306-SVH-21T-P1.1 | 18 |  |
| Ring crimp terminal | – | C20232480 | 548-BU-190540035 | 2 |  |
| 0.75 cm$^2$ wire (Brown) | H05VK075BR | – | 602-3055-100-07 | 0.4 m | |
| 0.75 cm$^2$ wire (Blue) | H05VK075BL | – | 602-3055-100-06 | 0.4 m | |
| 0.75 cm$^2$ wire (Red) | H05VK075RT | – | 602-3055-100-03 | 0.74 m | |
| 0.75 cm$^2$ wire (Black) | H05VK075SW | – | 602-3055-100-02 | 0.74 m | |
| 0.75 cm$^2$ wire (Yellow-Green) | H05VK075GN/GE | – | 602-3055-100-28 | 0.4 m | |
| 12 Vdc cooling fan | MF80251V3-1000U-G99 | – | – | 1 | |
| Polyamide cable tie; L: 100 mm; W: 2.5 mm | BMB1025 | – | – | 4 | |